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A MULTI-SENSOR LOCKSTEP SIMULATION ENVIRONMENT FOR DEVELOPING GNSS-DENIED UAS NAVIGATION

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ABSTRACT

Developing navigation for GNSS-denied urban flight needs an environment that can withhold GNSS on demand, expose every remaining sensor, and still provide exact ground truth. We present such an environment: a high-fidelity urban scene in Cosys-AirSim coupled in lockstep to an unmodified PX4 autopilot, where GNSS aiding is revoked *during flight* through a single estimator parameter while inertial, barometric, magnetic, ranging, and visual streams are logged against simulator truth at 20 Hz. Because the autopilot is the real flight stack and the coupling is MAVLink throughout, the same scenario definition drives software- and hardware-in-the-loop without modification. We demonstrate the environment with a repeatable inertial drift baseline, $|e| \approx 0.15t^{1.92}$ over 3 runs, and report the timing calibration required before such numbers can be trusted. The environment, not the baseline, is the contribution; absolute values are simulator-specific.

1 INTRODUCTION

Estimators that fuse inertial, visual, and ranging measurements in a factor-graph formulation (Dellaert and Kaess 2017) are a leading route to urban flight where GNSS is unreliable. Developing them is limited less by estimator theory than by obtaining synchronized multi-modal measurements during a controlled GNSS outage, with position truth accurate enough to grade the result. Flight tests supply realism but no truth and no repeatability; replayed public datasets supply neither a controllable outage nor a closed loop through a real autopilot.

This abstract contributes a simulation environment that closes that gap. Its properties are: (i) lockstep coupling between a photorealistic urban scene and an unmodified PX4 flight stack; (ii) GNSS revocation as a scenario variable applied in flight, leaving the sensor model untouched; (iii) synchronized logging of the remaining sensor suite against ground truth, in the form a factor-graph smoother consumes; and (iv) a MAVLink-only interface, so scenarios migrate to hardware-in-the-loop unchanged.

2 SIMULATION ENVIRONMENT

Vehicle dynamics, sensors, and the urban scene (Figure 1) come from Cosys-AirSim (Jansen, Verreycken, Schouten, Simon, Cornelis, and Hellinckx 2023). State estimation and control come from PX4 SITL, coupled over TCP in lockstep so simulator and autopilot never free-run relative to each other. A MAVSDK client commands the profile—arm, climb to 250 m, then a straight 250 m leg—and logs the EKF2 estimate against simulator truth.

GNSS revocation as a scenario variable. Outages clear the estimator's GNSS fusion mask at a configured ground-truth point, without restarting the autopilot or altering the simulated receiver, so only the estimator is perturbed. The height reference is moved to barometric before flight (reboot-scoped), because leaving it at its GNSS default removes every altitude source when GNSS is cut.

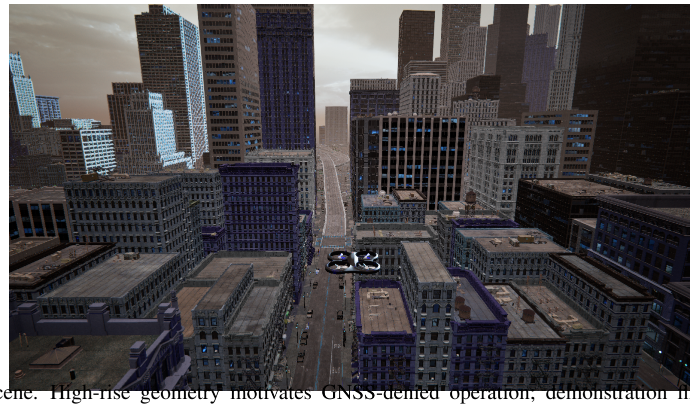

Figure 1: Simulated urban scene. High-rise geometry motivates GNSS-denied operation; demonstration flights use 250 m AGL so drift is not confounded by collisions.

Recorded streams. Each run logs local position, truth, raw inertial, barometric and magnetic measurements, and per-sweep LiDAR structure on one time base. Repetitions reuse a warm estimator that has reconverged with GNSS aiding before each outage; fusion is restored between flights so denied windows remain independent even when the autopilot process is not restarted.

3 DEMONSTRATION AND OBSERVED CALIBRATION

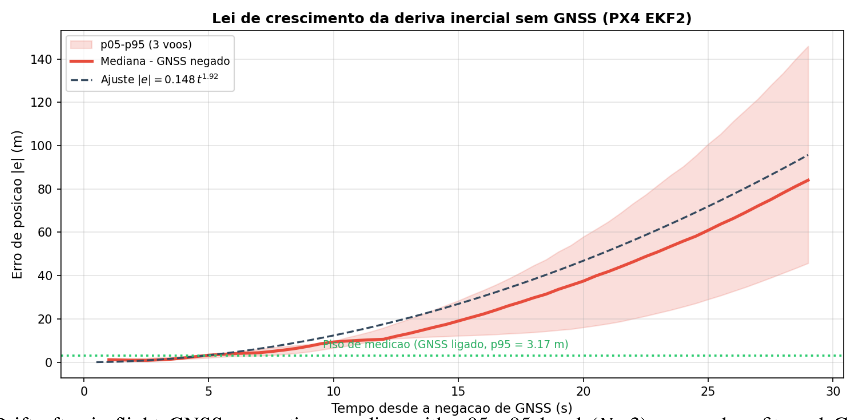

Figure 2: Drift after in-flight GNSS revocation: median with p05–p95 band (N=3), power-law fit, and GNSS-on floor.

Over 3 repetitions, position error during the outage grows as $|e| = 0.15\,t^{1.92}$ (median exponent; R^2 up to 0.96), reaching 86 m (median; 66–308) after about 33 s (Figure 2). An exponent near two is consistent with twice-integrated accelerometer bias. At 250 m AGL the LiDAR still returns structure in 59 % of sweeps (median 10 valid points), so ranging factors remain available even when flown clear of the dense canopy.

Estimate and truth are not sampled together: a 255 ms transport delay manufactured up to 3.4 m of velocity-proportional apparent error on GNSS-on flights. Correcting it leaves a residual floor of about 1.2 m median (3.2 m at p95) on the campaign baselines, which bounds the smallest resolvable drift. We report both because they condition the numbers above, not because they are the result.

4 CONCLUSIONS AND ONGOING WORK

The environment supplies controllable outages, synchronized sensing, exact truth, and repeatability with a real autopilot in the loop. Absolute drift reflects the simulator's inertial noise model rather than a specific airframe. Ongoing work replays the logged streams through a factor-graph smoother on the same outage windows; extends revocation toward receiver-level urban multipath as a further scenario variable; and moves the same scenarios to hardware-in-the-loop unchanged, because the interface is MAVLink on both sides.

REFERENCES

Dellaert, F., and M. Kaess. 2017. "Factor Graphs for Robot Perception". *Foundations and Trends in Robotics* 6(1–2):1–139.

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